

Vacancy-Loop Thermal Emission and Grain-Boundary Loop Truncation

An implementation plan for DisloCluster

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Abstract

Two changes to the immobile cluster-dynamics model are specified here in enough detail to be implemented directly. **(I)** Loop thermal emission is presently a dead constant — a placeholder $E_b = 5$ eV that appears in two files, is loaded by the 0-D solver and never used, and in MoDELlib3 suppresses the emission channel by 32 orders of magnitude. It is replaced by the size-dependent dissociation energies of Appendix D of the manuscript, which are transcribed into the code’s notation in §2, and the density and content equations are made mutually consistent in the three specific ways in which they currently are not. **(II)** A loop whose disc intersects the grain boundary is removed together with its stored defects; the number and content equations acquire the matching sink, and two new conservation accumulators book the defects that leave.

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1 Status of the code as found

1.1 Where loop emission lives today

Four independent treatments coexist, and no two of them agree.

Where	What it does	Size dependence
zerod/reaction_rates.py:74-76	$e_{v_vL} = \exp(-5/(k_B T))$, and $\phi_{v_emission_vL} = \omega_v(e_{v,vL} - e_v)$	none; dead — called nowhere
cpp_utils/parameters.h:91-96, 343-347	e_{v_vL} , $e_{sigma_v_vL}$, e_{v_iL} , $e_{sigma_v_iL}$, $e_{sigma_v_avL}$ are <i>required</i> CLI parameters, parsed into Params	never read by rate_ equations_core.h
rate_equations_core.h:155-159; zerod/rate_equations.py:314-341, 368-389	the only loop-loss channel that is not coalescence: $\dot{N} = -N/\tau_{vL}$, $\dot{c} = -n_{nuc}N/\tau_{vL}$, $\tau_{vL} = \tau_{vL}^0 e^{E_a/k_B T}$	none
ClusterDynamicsFEM.cpp:739-750 (in MoDELib3/src/ClusterDynamics/)	a genuine emission term, $\dot{c} = -S_k \bar{D}_v e^{-E_b/k_B T}$, with $E_b = Eb_eV[0] = 5.0$ eV from Zr3d_ghoniem.txt:237	perimeter only, through S_k

Three consequences follow, and each is worth stating on its own.

- 1. The coupled march has no loop emission at all.** The march's slow step is ZrMicro's (coupling/immobile.py), and ZrMicro never reads e_{v_vL} . The standalone 3-D run (useImmobileSolver=1) does have the channel. The two slow steps are therefore not the same model, which contradicts the coupling contract.
- 2. Even where the channel exists, 5 eV switches it off.** At $T = 573$ K, $k_B T = 0.04938$ eV, so $e^{-5/k_B T} = 1.05 \times 10^{-44}$. With Appendix D's $E_b^{v,(c)}$, which runs 1.26–1.47 eV over the entire size range the model visits (§2.7), the factor is 10^{-13} – 10^{-11} : the emission rate rises by **31 to 33 orders of magnitude**. A term suppressed by 10^{-44} is not an approximation, it is an absent term.

3. τ_{vL} is standing in for the missing physics. The two fitted parameters $\tau_{vL}^0 = 0.0237$ s and $E_a = 0.268$ eV (`input_data.py:188-189`) are the model’s only mechanism for dissolving a small vacancy loop. That is exactly what Appendix D describes from first principles, so after this change the two channels overlap and a decision must be made about τ_{vL} (§3.4).

1.2 Are the density and content equations consistent?

No, in three specific ways. The equations as written are

$$\dot{N}_{vL} = \nu_{\text{nuc}} - \frac{N_{vL}}{\tau_{vL}} - \Lambda_N^{\text{coal}}, \quad (1)$$

$$\dot{c}_{vL} = \underbrace{\phi_v - \phi_i}_{\text{net capture}} + G_{vL} - n_{\text{nuc}} \frac{N_{vL}}{\tau_{vL}} - \Lambda_c^{\text{coal}}. \quad (2)$$

(C1) The number and content sinks remove different loops. Equation (1) deletes loops at rate N/τ ; (2) removes n_{nuc} defects for each. But the mean content per loop is $\bar{n} = c/N$, and on the 500 nm matched leg $\bar{n}_c = 7.4 \times 10^4$ against $n_{\text{nuc}} = 50$ — three orders of magnitude apart. The channel therefore deletes loops while leaving 99.93% of their defects behind in c_{vL} , and the surviving population’s mean size $\bar{n}$ rises because loops were deleted, not because anything was absorbed. This is a deliberate “embryo dissolution” reading of τ_{vL} , documented at `reaction_rates.py:641-649` and mirrored verbatim at `ClusterDynamicsFEM.cpp:726-737`, and it is self-consistent only in the limit where every dissolving loop really is at its birth size. It is not self-consistent as an emission model, and it does not satisfy $\dot{c} = \bar{n} \dot{N}$.

(C2) There is no channel by which a loop shrinks by emitting. A $\langle c \rangle$ loop can currently only shrink by absorbing an interstitial (`vL_i`), which is booked as recombination in `cum_recomb` (`rate_equations_core.h:416`). Emission differs in every respect that matters to the bookkeeping: it removes content, returns a *vacancy* to the free pool, is not a Frenkel event, and does not remove a loop until the loop is gone. None of that exists in the 0-D, and the corresponding SIA channel for $\langle a \rangle$ loops (Appendix D, Eq. (D.12)) exists in neither code.

(C3) The 0-D has no critical size for $\langle c \rangle$ loops. The minimum-stable-size gate `size_gate(r) / r_min_a` is applied to the $\langle a \rangle$ families only (`rate_equations_core.h:321-322`, and the `if(!is_vac[k])` guard at line 249). The $\langle c \rangle$ family has no lower size bound at all; nothing stops $\bar{n}_c$ from falling below one vacancy. Emission supplies exactly this bound, and supplies it as physics rather than as a gate: N^* is where $C_v = C_v^e(N)$.

1.3 Where the grain boundary truncates loops today

Nowhere, in the solve. The geometric refusal exists only in post-processing and in the discrete handoff: `coupling/transition.py:456` (`fits_in_crystal`) predicts which loops `microstructureGenerator` will reject, and `studies/hardening.py` inherits it through `cube_population`. The continuum march itself lets a loop of any radius sit at a node one Burgers vector from a Dirichlet face.

The measured consequences are already recorded in `CLAUDE.md` as artifacts to be worked around rather than as a missing term:

- on the 1 μm single crystal, 6522 of 24115 CD nodes lie on the six Dirichlet faces and at 21 dpa carry **99.8%** of the total $\langle a \rangle$ density, with $c_a/N_a = 300.8$ against $n_{\text{iL,nuc}} = 300$ — every loop there is a fresh nucleus;
- the domain mean of any loop quantity is consequently unusable, and the documentation prescribes the interior-quartile mean everywhere instead (`post/report.py`, `studies/hardening.py`, `post/discrete_loops.py region="interior"`);
- the $\langle c \rangle$ handoff at 500 nm keeps only 0.96% of loops at 1 dpa, measured, because the continuum holds loops that cannot geometrically fit.

The stated cause is correct — “the boundary is a sink for mobile defects but not for loops” — and it is a *missing sink*, not an inevitable artifact. Supplying it is item (II), and the near-boundary divergence is the validation target that decides whether it worked.

2 Appendix D in the code’s notation

This section transcribes Appendix D (“Dissociation”) of the manuscript. The equation numbers (D.1)–(D.13) are local to this document; the physics, the parameter table and the implementation recipe are the manuscript’s. §2.6 reconciles its parameters against the values the code carries, and §2.8 lists four points to settle with the author before implementation — one of them blocking.

2.1 Chemical potential and binding energy

The bulk equilibrium vacancy concentration is

$$C_v^0 = \exp\left(-\frac{E_v^f}{k_{\text{B}}T}\right). \quad (\text{D.1})$$

At the surface of a *vacancy* cluster of size α it is raised, and at the surface of an *interstitial* cluster of size β lowered,

$$C_v^e(\alpha) = C_v^0 e^{\mu(\alpha)/k_{\text{B}}T}, \quad C_v^e(\beta) = C_v^0 e^{-\mu(\beta)/k_{\text{B}}T}, \quad (\text{D.2–D.3})$$

so that with the binding energies

$$E_b^{(v)}(\alpha) = E_v^f - \mu(\alpha), \quad E_b^{(i)}(\beta) = E_v^f + \mu(\beta), \quad (\text{D.4–D.5})$$

both cases collapse to the single statement

$$\boxed{C_v^e = \exp\left(-\frac{E_b}{k_{\text{B}}T}\right)}. \quad (\text{D.6})$$

Equation (D.6) is the object the rate equations need, and it is exactly what the original author’s dead skeleton was reaching for: `phi_v_emission_vL` = $\omega_v(e_{v,vL} - e_v)$ with $e_v = e^{-E_v^f/k_{\text{B}}T}$ is the difference between C_v^e and C_v^0 . This plan restores the intended structure and supplies the missing size dependence.

In hcp Zr neither cluster type is a 3-D void: the vacancy population collapses onto $\{0001\}$ as faulted $\langle c \rangle$ loops with $\mathbf{b} = \frac{1}{6}\langle 20\bar{2}3 \rangle$, and the interstitial population assembles into prismatic $\langle a \rangle$ loops with $\mathbf{b} = \frac{1}{3}\langle 11\bar{2}0 \rangle$ on $\{10\bar{1}0\}$. The remainder of the appendix is the dissociation kinetics of those two families.

2.2 Loop geometry and the perimeter chemical potential

A circular loop of N point defects with Burgers vector b occupies habit area $A_N = N\Omega/b$, hence

$$R_L(N) = \sqrt{\frac{A_N}{\pi}} = \sqrt{\frac{N\Omega}{\pi b}}. \quad (\text{D.7})$$

This is already the code's radius. `rate_equations_core.h:82` forms $r = \ell\sqrt{c/N}$ with $\ell = \sqrt{\Omega/\pi b}$ (`input_data.py:366, 378`), and $\sqrt{c/N} = \sqrt{\bar{n}}$, so $r \equiv R_L(\bar{n})$ identically. No new geometry code is needed.

The defect chemical potential at the loop perimeter — self-stress, fault energy, and applied stress — is

$$\mu_L(N) = \frac{\Omega}{b} \left[\frac{\mu_s b^2}{4\pi(1-\nu)R_L(N)} \ln \frac{R_L(N)}{r_0} + \gamma_{\text{SF}} - \sigma_{\text{app}} b \right], \quad (\text{D.8})$$

with μ_s the shear modulus, ν Poisson's ratio, $r_0 \simeq b$ the core cutoff, γ_{SF} the fault energy of the habit plane (zero for a perfect loop), and σ_{app} the applied stress resolved on the loop normal.

Dimensional note. As printed in the manuscript the last term is $-\sigma_{\text{app}}$, not $-\sigma_{\text{app}}b$. $(\Omega/b)\sigma_{\text{app}}$ has units $\text{m}^2 \text{Pa} = \text{J m}^{-1}$, not joules, whereas the other two bracket terms give energies; carrying b makes the term $-\sigma_{\text{app}}\Omega$, the standard climb work per defect. (D.8) is written here in the dimensionally consistent form. See §2.8, item 2.

2.3 The two families

For the SIA-type prismatic $\langle a \rangle$ loop, with $b = b_a$ and $\gamma_{\text{SF}}^{(a)}$,

$$E_b^{i,(a)}(n) = E_i^f + \mu_L^{(a)}(n), \quad (\text{D.9})$$

and for the vacancy-type basal $\langle c \rangle$ loop, with $b = b_c$ and $\gamma_{\text{SF}}^{(c)}$,

$$E_b^{v,(c)}(m) = E_v^f - \mu_L^{(c)}(m). \quad (\text{D.10})$$

The $-\mu_L$ in (D.10) reflects the vacancy nature of the $\langle c \rangle$ loop: vacancies are bound because removing one from the perimeter requires inflating the faulted region. Cross-channel emission — an SIA from a vacancy $\langle c \rangle$ loop, a vacancy from an $\langle a \rangle$ loop — is energetically suppressed and is omitted from the master equations, which matches the structure of the code: each family exchanges only its own defect type through the emission channel.

2.4 Small-cluster blend

The continuum form (D.8) holds for $N \gtrsim N_{\text{cont}} = 20$ and fails below it, where the discreteness of the aggregate dominates. For $N \leq N_{\text{small}} = 5$ the binding energies are tabulated from DFT; between the two, the same exponential correction used for voids is applied,

$$E_b^{\text{mod}}(N) = E_b^{\text{cont}}(N) + [E_b^{\text{DFT}}(2) - E_b^{\text{cont}}(2)] e^{-\lambda(N-2)}, \quad \lambda = \frac{\ln 100}{8} \simeq 0.575, \quad (\text{D.11})$$

the decay constant chosen so that the correction is $\sim 1\%$ at $N = 10$. The evaluation recipe is: tabulated DFT for $N \leq 5$; (D.11) for $5 < N < 20$; pure (D.8) for $N \geq 20$.

N	2	3	4	5
$E_b^{i,(a)}(N)$ [eV], di-SIA $\langle a \rangle$ configuration	0.80	1.05	1.30	1.45
$E_b^{v,(c)}(N)$ [eV], divacancy basal configuration	0.20	0.45	0.70	0.85

The blend is inert in the model as it stands, and that is worth knowing before implementing it. Loops are born at $n_{\text{vL},\text{nuc}} = 50$ and $n_{\text{iL},\text{nuc}} = 300$, both above $N_{\text{cont}} = 20$, and the measured mean sizes run from $\bar{n}_a \simeq 300$ to $\bar{n}_c \simeq 7.4 \times 10^4$. The $N < 20$ branch can therefore only be reached if emission drives a family down toward dissolution — which is precisely the regime this change creates. Implement it, but expect it to be exercised only near dissolution, and treat any run that spends time there as a signal in its own right.

2.5 Emission rates, and how they map onto the code

$$\varepsilon_i^{(a)}(n) = A_{\text{loop}} (n-1)^{1/2} Z_i^{(a)} \omega_i^{\text{eff}} \exp\left(-\frac{E_b^{i,(a)}(n)}{k_{\text{B}}T}\right), \quad (\text{D.12})$$

$$\varepsilon_v^{(c)}(m) = A_{\text{loop}}^{(c)} (m-1)^{1/2} Z_v^{(c)} \omega_v^{\text{eff}} \exp\left(-\frac{E_b^{v,(c)}(m)}{k_{\text{B}}T}\right). \quad (\text{D.13})$$

The $(N-1)^{1/2}$ prefactor is the code’s S_k , already built. (D.12)–(D.13) are per-loop rates; the master equations need the rate per unit volume, which is εN_k . Since $N^{1/2} \propto R_L$ by (D.7),

$$\varepsilon_k N_k \propto N_k \sqrt{\bar{n}_k} = \sqrt{N_k c_k} \propto S_k, \quad S_k = \frac{\ell_c}{\ell} s_k \sqrt{N_k c_k},$$

where S_k is the sink strength the code assembles as `pref_vL` in `rate_equations_core.h`. The population-level emission is therefore

$$\mathcal{E}_k = S_k \bar{D}_m Z_k \exp\left(-\frac{E_b(\bar{n}_k)}{k_{\text{B}}T}\right) = S_k \bar{D}_m Z_k C^e(\bar{n}_k), \quad (3)$$

which is *exactly* MoDELlib3’s existing expression at `ClusterDynamicsFEM.cpp:745` with the constant $e^{-E_b/k_{\text{B}}T}$ replaced by the size-dependent (D.6). ω^{eff} maps onto $\bar{D}_m$ (the code’s `omega_m = (\det D_m)^{1/3}`), Z onto the existing capture efficiencies, and A_{loop} is absorbed into s_k (`loopSinkScale`). The $(N-1)$ against N is a one-defect correction, negligible at $\bar{n} \geq 50$ and subsumed by the blend below it.

The only genuinely new quantity is $E_b(\bar{n}_k)$. Everything else in (3) is already assembled and already validated.

2.6 Parameters, and reconciliation with the code

Quantity	App. D	Code	Action
$\gamma_{\text{SF}}^{(c)}, \{0001\}$	200 mJ/m ²	213 mJ/m ² (ISF_SI)	Already present. Agree to 6%; adopt Appendix D’s 200 or keep ISF_SI — state which

Quantity	App. D	Code	Action
$\gamma_{\text{SF}}^{(a)}, \{10\bar{1}0\}$	130 mJ/m ²	128 mJ/m ² (gamma_sf)	Already present, and this resolves an apparent conflict: the 0-D's gamma_sf and the material file's ISF_SI are not two values of one fault, they are the two faults. Label them
μ_s	35.5 GPa	33 GPa (mu0_SI)	Reconcile (7.6%); μ_s enters μ_L linearly
ν	0.34	0.34	agree
b_a	0.3232 nm	0.323 nm (b_SI)	agree to 0.06%; the code's value is deliberate, so that ZrMicro and MoDELib3 share one a
b_c	0.6173 nm, stated as $\frac{1}{6}\sqrt{3a_0^2 + 4c_0^2}$	0.2575 nm, = $c/2$	The largest single discrepancy — see below
r_0	$\simeq b$	—	new; add per family
E_v^f, E_i^f	—	1.6, 3.0 eV	already present, input_data.py:129–130
DFT $E_b(2 \dots 5)$	tabulated, §2.4	—	new: two 4-entry tables
$\lambda, N_{\text{small}}, N_{\text{cont}}$	0.575, 5, 20	—	new

The $\langle c \rangle$ Burgers vector: three values, and they must be reduced to one. Appendix D's table gives $b_c = 0.6173$ nm and its text gives $\mathbf{b} = \frac{1}{6}\langle 20\bar{2}3 \rangle$. These do not agree with each other, and neither agrees with the code:

Source	$ b_c $	Note
Appendix D table	0.6173 nm	does not follow from the printed formula
Appendix D formula $\frac{1}{6}\sqrt{3a_0^2 + 4c_0^2}$	0.1955 nm	evaluated at $a_0 = 0.3232$, $c_0 = 0.5153$ nm
$ \frac{1}{6}\langle 20\bar{2}3 \rangle = \sqrt{a_0^2/3 + c_0^2/4}$	0.3181 nm	the magnitude of the stated vector
the code, everywhere	0.2575 nm	= $c/2$, a perfect $\frac{1}{2}[0001]$ loop

This is not a detail. b_c enters R_L as $b^{-1/2}$ and μ_L as Ω/b times terms carrying b and b^2 , and it also sets ℓ_c , the loop sink strength, and the continuum→discrete transfer. Two further points bear on the choice:

- **The code was very recently standardized on $c/2$** , in four places that must stay equal (HEXlattice.cpp, Zr3d_ghoniem.txt immobileSpeciesBurgers, post/discrete_loops.py, zerod/input_data.py), which bought $f_{\text{burgers}} = \sqrt{b_{\text{cd}}/b_{\text{dd}}} = 1.000000$ at the transfer and made the $\langle c \rangle$ part of the fit stale. Moving to a faulted $\frac{1}{6}\langle 20\bar{2}3 \rangle$ loop would move all four again, and would make the loop faulted where the code currently treats it as perfect — at which point $\gamma_{\text{SF}}^{(c)}$ becomes load-bearing rather than decorative.
- $b_c = 0.6173$ nm **breaks the continuum branch over part of the range in which it is declared valid.** With $r_0 = b$, $\ln(R_L/r_0) < 0$ for $N < \pi b^3/\Omega$, which is $N < 31.8$ at $b_c = 0.6173$ nm — above $N_{\text{cont}} = 20$, so μ_L is negative exactly where (D.8) is supposed to

hold ($\mu_L = -0.134$ eV at $N = 20$, giving $E_b = 1.73$ eV, *higher* than the bulk E_v^f). The same threshold is $N = 2.3$ at $b_c = c/2$ and $N = 4.3$ at $b_c = 0.318$ nm, where it is harmless. Whichever b_c is adopted, N_{cont} must satisfy $N_{\text{cont}} > \pi b^3/\Omega$.

2.7 Numbers

Evaluated from (D.7)–(D.10) at $\mu_s = 35.5$ GPa, $\nu = 0.34$, $r_0 = b$, $\Omega = 2.326553 \times 10^{-29}$ m³, $\sigma_{\text{app}} = 0$, $T = 573$ K ($k_B T = 0.04938$ eV). Sizes span the model’s actual range: nucleation ($n_{\text{vL,nuc}} = 50$, $n_{\text{iL,nuc}} = 300$) up to the measured 500 nm interior means.

N	R_L [nm]	μ_L [eV]	E_b [eV]	$C^e = e^{-E_b/k_B T}$
$\langle c \rangle$ vacancy loop, $b_c = c/2 = 0.2575$ nm (the code’s value), $\gamma_{\text{SF}}^{(c)} = 200$ mJ/m ² , $E_v^f = 1.6$ eV				
20	0.76	0.3407	1.2593	8.4×10^{-12}
50	1.20	0.3181	1.2819	5.3×10^{-12}
300	2.94	0.2454	1.3546	1.2×10^{-12}
5000	11.99	0.1641	1.4359	2.3×10^{-13}
74281	46.22	0.1308	1.4692	1.2×10^{-13}
$\langle a \rangle$ interstitial loop, $b_a = 0.3232$ nm, $\gamma_{\text{SF}}^{(a)} = 0$ (perfect loop), $E_i^f = 3.0$ eV, using $E_b = E_i^f - \mu_L$ — see §2.8 item 1				
20	0.68	0.2194	2.7806	3.5×10^{-25}
300	2.62	0.1604	2.8396	1.1×10^{-25}
74281	41.26	0.0236	2.9764	6.6×10^{-27}
bulk vacancy, $E_v^f = 1.6$ eV			1.600	8.4×10^{-15}
current placeholder			5.000	1.1×10^{-44}

Four readings of this table matter.

1. **The placeholder sits 10^{31} – 10^{33} below anything physical.** The present code has no emission channel however the term is wired.
2. C_v^e is 14–1000× **the bulk equilibrium but far below the irradiation-driven C_v over most of the dose range**, so emission is expected to be a *correction* to $\langle c \rangle$ growth at large N and the *dominant* loss near dissolution. The change should reshape the small end of the size distribution and barely move the large end. A change concentrated at large R means $\gamma_{\text{SF}}^{(c)}$ or b_c is wrong.
3. **The $\langle a \rangle$ SIA channel is inert at 573 K.** $E_b^{i,(a)}$ is pinned near $E_i^f = 3.0$ eV because $\mu_L^{(a)} \leq 0.22$ eV, giving 10^{-25} — thirteen orders below the $\langle c \rangle$ channel. Implement it for completeness and for higher temperatures, but do not expect it to move any measured quantity, and do not let it hold up item (I).
4. **The size dependence is weak, spanning 0.21 eV over $N = 20 \dots 7 \times 10^4$** , i.e. only $\sim 70\times$ in rate across the model’s whole range. The dominant effect of this change is therefore *switching the channel on*, not its size dependence — which means the γ_{SF} and b_c choices, which shift the whole curve, matter more than the elastic term, which sets its slope.

2.8 Four points to settle with the author

1. **The sign in (D.9).** As printed, $E_b^{i,(a)} = E_i^f + \mu_L$, so E_b *rises* as the loop shrinks (3.22 eV at $N = 20$ against 3.02 eV at $N = 7 \times 10^4$) and a small $\langle a \rangle$ loop emits *less* — the opposite

of (D.10)’s behavior for $\langle c \rangle$, and the opposite of Gibbs–Thomson. The $+$ appears to have carried over from (D.5), which is the *cross-species* case (a vacancy at an interstitial cluster). The like-species form consistent with (D.4) is $E_b^{i,(a)} = E_i^f - \mu_L^{(a)}$, which is what §2.7 uses. The detailed-balance test of §5 distinguishes the two unambiguously. *Non-blocking* — the channel is inert at 573 K either way.

2. **The σ_{app} term in (D.8)** is dimensionally inconsistent as printed; $-\sigma_{\text{app}}b$ inside the bracket ($= -\sigma_{\text{app}}\Omega$ overall) is assumed here. Note also that this convention gives $C_v^e \propto e^{-\sigma\Omega/k_{\text{B}}T}$, the *opposite* sign to the existing `update_stress_effects` (`reaction_rates.py:104`, $e^{+\sigma_h\Omega/k_{\text{B}}T}$). That code is dead, so nothing conflicts today, but the convention must be stated once and used everywhere — it is the entry point for SIPA and for the hoop-stress coupling.
3. b_c : three values, §2.6. *Blocking*, and it is also a decision about whether the $\langle c \rangle$ loop is perfect ($\frac{1}{2}[0001]$, as the code has it) or faulted ($\frac{1}{6}\langle 20\bar{2}3 \rangle$, as the appendix text has it).
4. $\gamma_{\text{SF}}^{(a)}$: the text says the perfect $\frac{1}{3}\langle 11\bar{2}0 \rangle$ loop is unfaulted ($\gamma_{\text{SF}}^{(a)} = 0$) while the table lists 130 mJ/m² on $\{10\bar{1}0\}$. *Non-blocking*: it moves $E_b^{i,(a)}$ by 0.06 eV in an already inert channel.

One deliberate difference from an independent derivation, recorded so it is not mistaken for an error. (D.8) is the *capillary* chemical potential $\propto \Gamma(R)/R$, not the exact derivative dE_L/dN of the loop energy $E_L = 2\pi R\Gamma(R) + \pi R^2\gamma_{\text{SF}}$. The two differ by $\Omega\mu_s b/[4\pi(1-\nu)R]$ — the derivative of the logarithm — which is 0.054 eV at $N = 300$, a factor of 3.0 in the emission rate. The appendix’s form is what is implemented, because it is the manuscript’s; the difference is noted because a reader deriving it independently will arrive at the other one.

3 Item (I): implementing size-dependent emission

3.1 Where the term goes

Emission is written as an additive content sink rather than folded into the capture term. The two are algebraically identical — writing $\phi_v = S_k Z_v^c \bar{D}_v (C_v - C_v^e)$ and splitting it gives the same thing — but the additive form leaves the legacy expressions *character for character* intact when the channel is off, which this repository requires: `CLAUDE.md` records that regrouping `growth + nuc + G` into `growth + (nuc + G)` moved the tenth significant digit and broke the bit-identity of `loop_model = 0` (sha256 `ff2980c4` against `8bcc7780`).

For each family k , with $\mathcal{E}_k$ from (3) — the vacancy species for the $\langle c \rangle$ slot, the interstitial species for the $\langle a \rangle$ slots:

$$\dot{c}_k = -\mathcal{E}_k, \tag{4}$$

$$\dot{C}_{v \text{ or } i} = +\mathcal{E}_k, \tag{5}$$

$$\dot{N}_k = -\frac{\mathcal{E}_k}{\bar{n}_k} g_{\text{dis}}(\bar{n}_k), \quad \bar{n}_k = \frac{c_k}{N_k}, \tag{6}$$

with g_{dis} a C^1 smoothstep equal to 1 at $\bar{n}_k = n_{\text{min}}$ and 0 for $\bar{n}_k \geq (1+w)n_{\text{min}}$ — the same $x^2(3-2x)$ the code already uses for `size_gate`. n_{min} is naturally $N_{\text{small}} = 5$, the bottom of the DFT table.

3.2 Why (6) closes the number balance

Emission removes content, not loops — until a loop has nothing left. For $\bar{n}_k \gg n_{\text{min}}$ every emitted defect comes off a surviving loop and $\dot{N} = 0$; as $\bar{n}_k \rightarrow n_{\text{min}}$ every emitted defect is the last one

of some loop, and $n_{\min}$ of them delete one loop, giving $\dot{N} = -\mathcal{E}/\bar{n}$. This is the minimal closure consistent with a single-moment (N, c) description, and it repairs defect (C1) of §1.2: on this channel $\dot{c} = \bar{n} \dot{N}$ holds identically in the dissolution limit, which the $n_{\text{nuc}} N/\tau$ channel satisfies nowhere.

3.3 The accumulators need no new channel for item (I)

Emission is an *internal transfer*, loop content $\rightarrow$ free pool, exactly like `ann_cont_vL`. $V_{\text{stored}} = C_v + c_{vL} + c_{avL}$ and $I_{\text{stored}} = C_i + 2C_{2i} + 3C_{3i} + c_{iL} + c_{aiL}$ are unchanged by it, so `cum_prod_*`, `cum_recomb_*` and `cum_sink_*` are untouched and the closure identity still holds. **It must not be added to `cum_recomb`** — that would double-count, because no defect of the opposite type is destroyed.

3.4 What happens to τ_{vL}

τ_{vL} and Appendix D describe the same physics. Keeping both double-counts embryo dissolution; removing τ_{vL} invalidates two of the 28 fitted parameters. The resolution is a switch, defaulting to present behavior, so that nothing already published moves:

emission_model	Channels	Status
0	τ_{vL} only	default ; bit-identical to today
1	τ_{vL} + emission	staged intermediate; double-counts near dissolution, useful only to size the emission term against the existing one
2	emission only, τ_{vL} disabled	the target model; two fewer fitted parameters, requires a refit

This is the fourth “the fit is stale” item in this code base — alongside Ω , the $\langle c \rangle$ Burgers vector and `loop_model` — and they compound. If query 3 of §2.8 moves b_c as well, it is the fifth, and it touches the same four places the last b_c change did. Mode 2 must not be compared with experiment before the refit.

3.5 Files to change

File	Change
zerod/input_data.py	add <code>mu_s</code> , <code>nu</code> , <code>r0_a</code> , <code>r0_c</code> , the two DFT tables, λ , N_{small} , N_{cont} ; label the two fault energies — <code>gamma_sf</code> (0.128 J/m ²) is $\gamma_{\text{SF}}^{(a)}$ and <code>ISF_SI</code> (0.213) is $\gamma_{\text{SF}}^{(c)}$; they were never in conflict. Derive $n_{\min}$ per family. The 0-D currently carries no shear modulus at all
zerod/reaction_rates.py:55–76	delete the 5 eV constants; add <code>mu_L(N, fam)</code> , <code>Eb_loop(N, fam)</code> (with the (D.11) blend) and <code>C_eq_loop(N, fam)</code> implementing (D.6)–(D.11); make <code>phi_v_emission_vL</code> live and size-dependent; add <code>emission_k</code> and <code>dissolution_num_k</code> for all four slots
zerod/rate_equations.py	wire (4)–(6) into <code>dCv_dt</code> , <code>dCi_dt</code> (the $\langle a \rangle$ channel), and all eight loop equations (lines 203–292 and 294–389)

File	Change
cpp_utils/parameters.h	replace the five dead <code>e_*</code> scalars (lines 87–96, 339–347) with the Appendix D parameter set and <code>emission_model</code> ; keep the old keys accepted-and-ignored for one release so existing batch files still parse
cpp_utils/rate_equations_core.h	<code>mu_L/Eb_loop/C_eq_loop</code> helpers on the AD type; emission enters through the shared family slots $f_{\text{annc}}/f_{\text{annn}}$, which both loop models already use, so <i>one</i> implementation serves both. The <code>loop_model == 0</code> verbatim block gains exactly two subtractions per family, guarded by <code>if (P.emission_model)</code> . The blend makes E_b piecewise — keep it C^1 across $N = 5$ and $N = 20$ or the AD Jacobian is discontinuous
zerod/cpp_bridge.py:72–80	stop emitting the dead parameters; emit the new ones
ClusterDynamicsFEM.cpp:739–750	replace <code>cdp.Eb(0)</code> with $E_b(\bar{n}_k)$; the implicit-update structure is unchanged, because the term is still first order in c after the existing division. Extend from the vacancy family to all four
Library/Materials/Zr3d_ghoniem.txt	<code>Eb_eV</code> retains its <i>cluster</i> entries (indices 2, 3); the loop entries become unused — annotate rather than silently leave 5.0 there. Add μ_s if it is to differ from <code>mu0_SI</code> , and the $\{0001\}/\{10\bar{1}0\}$ fault energies as a labelled pair

4 Item (II): removing loops that intersect the grain boundary

4.1 The geometric criterion

A circular loop of radius R and habit normal $\hat{\mathbf{n}}$ centred at $\mathbf{x}$ has half-extent

$$h(\hat{\mathbf{n}}, \hat{\mathbf{u}}) = R\sqrt{1 - (\hat{\mathbf{n}} \cdot \hat{\mathbf{u}})^2}$$

along $\hat{\mathbf{u}}$. It therefore intersects the face of outward normal $\hat{\mathbf{u}}_f$ and offset b_f — the convex-hull representation `fields.domain_faces` already returns, as $\mathbf{N}\mathbf{x} \leq \mathbf{b}$ — whenever

$$d_f(\mathbf{x}) \equiv b_f - \hat{\mathbf{u}}_f \cdot \mathbf{x} < R_k\sqrt{1 - (\hat{\mathbf{n}}_k \cdot \hat{\mathbf{u}}_f)^2}. \quad (7)$$

Two pieces are already in place and must be reused rather than re-derived: `post/fields.py:140` `gb_distance` gives $\min_f d_f$ for any convex domain (hexagonal prism as well as cube), and `coupling/transition.py:456` `fits_in_crystal` applies the same test vertex by vertex to predict `microstructureGenerator`’s decision. After this change those two must agree: the continuum will no longer hold loops the generator would refuse.

Periodic faces are not boundaries. `BOUNDARY['periodic_face_ids']` must gate (7) face by face. A fully periodic cell — the hardening cube of `studies/hardening.py` — must be bit-identical after this change.

4.2 From a step to a rate

The model carries one mean size per family per node, so (7) is a step in R and would make the ODE discontinuous. Regularize with the physical fact that loop centres are distributed within the

node’s control volume: with local nodal spacing $w(\mathbf{x})$,

$$\varphi_k^{\text{GB}}(\mathbf{x}) = \text{clamp}\left(\frac{R_k^{\text{eff}}(\mathbf{x}) - d(\mathbf{x}) + w/2}{w}, 0, 1\right), \quad R_k^{\text{eff}} = R_k \sqrt{1 - (\hat{\mathbf{n}}_k \cdot \hat{\mathbf{u}}_{f^*})^2},$$

f^* being the nearest non-periodic face. φ_k^{GB} is the fraction of that node’s loops whose centres already lie within reach of the surface. The removal rate is the rate at which the population is swept through the ramp:

$$\Gamma_k^{\text{GB}} = \varphi_k^{\text{GB}} \left[\frac{\max(\dot{R}_k, 0)}{w} + \frac{1}{\tau_{\text{GB}}} \right], \quad \dot{N}_k = \Gamma_k^{\text{GB}} N_k, \quad \dot{c}_k = \Gamma_k^{\text{GB}} c_k. \quad (8)$$

The floor $1/\tau_{\text{GB}}$ (default: one substep) drains a population that is already intersecting but no longer growing; without it a node whose loops stop growing keeps them forever.

The content sink uses the MEAN content, and that is the point. $\dot{c}_k = \bar{n}_k \dot{N}_k$ holds exactly, because a boundary-truncated loop leaves *whole*. This is the opposite of the embryo channel of §1.2, where the birth content is the right choice, and of the emission channel of (6), which removes content first and loops only at the end. The contrast deserves a comment in the source: three channels remove loops from three different parts of the size distribution and none of them may share a formula.

$\dot{R}_k$ is already available. The coalescence lambda at `rate_equations_core.h:441` computes `drdt = lscale*gcont/(2*sqrt(Cnum*Ccont))` from the gain-side absorption. Reuse it rather than differentiating R a second time — and note that once item (I) is in, $\dot{R}_k$ acquires an emission contribution, so the two items are not fully independent at this one point.

4.3 Conservation: two new accumulators

Defects that leave with a truncated loop leave the crystal. They are neither recombination nor network-sink absorption, and they must not be allowed to fall into the published `grain_boundary` channel, which is defined as the closure residual and is independently cross-checked against the *mobile* flux by `post/boundary_flux.py`. Letting loop removal land there would corrupt an independent measurement.

Add two accumulators **after** `rho_N`, so that `idx_rhoN = 18` is unchanged and no existing index shifts:

Index	Name	Integrand
19	<code>cum_gb_i</code>	$\sum_{k \in \langle a \rangle} \Gamma_k^{\text{GB}} c_k$ — interstitial atoms leaving with truncated $\langle a \rangle$ loops
20	<code>cum_gb_v</code>	$\sum_{k \in \langle c \rangle} \Gamma_k^{\text{GB}} c_k$ — vacancy atoms leaving with truncated $\langle c \rangle$ loops

The closure identity in `zerod/post_process.py:251-300` becomes

$$(\text{stored} - \text{stored}_0) = \text{prod} - \text{recomb} - \text{sink} - \text{gb_loop} + \text{residual},$$

and `_calculate_conservation` publishes `grain_boundary_loops` (measured, from the accumulator) separately from `grain_boundary` ($= -\text{residual}$, the mobile-species flux). The `*_fractions` figures gain one channel and their “Sum ($= 1$)” line must be re-checked.

4.4 Getting the geometry into a 0-D ODE

The slow step is a point model with no notion of position. Two routes:

G1 (recommended) — per-case CLI parameters. Pass d_{gb} and w (both in b) as two extra tokens per node through the existing base+delta batch protocol (`zerod/cpp_bridge.py:513, _batch_lines`), computed once per node in `coupling/immobile.py` from `fields.gb_distance(nodes, faces=...)`. The removal term then lives in `rate_equations_core.h` beside coalescence and is integrated implicitly with everything else. The two tokens differ per node, so they become deltas rather than base entries; this costs two numbers per case line and **reduces dedup efficiency** (currently $\times 1.25$), because formerly identical boundary-shell nodes now differ. Measure the slow-step cost before and after.

G2 — an operator-split projection. Apply the removal in `coupling/march.py` between substeps. Simpler, and needs no C++ change, but it adds a second splitting error and cannot see R_k crossing d *inside* a substep, which is exactly when truncation happens.

Absent d_{gb} the channel is off, so the standalone 0-D and every existing batch file stay bit-identical.

The 3-D immobile solver needs the same term. `ClusterDynamicsFEM.cpp`'s `solveImmobileClusters` knows the nodal position directly, so (8) is a few lines there and needs no parameter plumbing. Both slow steps must carry it, or the two routes stop being the same model — the failure this plan opens with.

4.5 State-length compatibility

The native state grows $19 \rightarrow 21$. Each of these hard-codes the old length and must be audited:

Location	What it assumes
<code>coupling/immobile.py:98</code>	<code>N_EQ = 19</code>
<code>coupling/field.py:297</code>	<code>if Y.shape[1] != 19: raise</code>
<code>coupling/seeded_march.py:85</code>	<code>y[:19]</code> , zero-padded to 19
<code>coupling/checkpoint.py</code>	documented 19-column state
<code>post/fields.py:677, post/march_report.py:193</code>	19-variable form
<code>cpp_utils/rate_equations.cpp</code>	the index-18 contract for <code>rho_N</code>
<code>post/movies.py, driver.py:156</code>	19-column reads

Follow the precedent already in `zerod/reaction_rates.py:670-690` (`current_rho_N` tolerates 12- and 18-length vectors): make every reader length-tolerant rather than switching them all at once, so an existing `march_state.npz` still loads. Record the state length in `summary.json` beside `loop_model`, for the same reason — **the array does not say which layout it is in**.

5 Validation

5.1 Regression — nothing may move by default

1. `emission_model = 0` and no $d_{gb} \Rightarrow$ the `loop_model` 0 25-point integration hash stays `8bcc7780`. This is the gate, and it is why the legacy assembly must not be tidied.

2. The fully periodic hardening cube is unchanged in every measured quantity ($\Delta\tau$, $n_{\text{obstacles}}$).
3. `studies/hardening verify` and the existing conservation checks still pass.

5.2 Item (I) — targeted

1. **Detailed balance.** With $G = 0$ and C_v pinned at $C_v^e(N_0)$, a loop of size N_0 must be stationary to solver tolerance. This is a unit test, and it is the one test that catches a sign error in (D.9)/(D.10) — see §2.8 item 1.
2. **Reproduce §2.7.** A table test on μ_L , E_b and C^e at the tabulated sizes, plus continuity of the blend across $N = 5$ and $N = 20$ and of its first derivative.
3. **The 10^{32} claim.** Print $\mathcal{E}_k$ at 573 K under the old constant and under $E_b(\bar{n})$ on the same state, and confirm the ratio.
4. **Shape, not scale.** Emission should reshape the small end of the $\langle c \rangle$ size distribution (`gb/size_dist_c_*.png`) and leave the large end within a few percent. A change concentrated at large R means $\gamma_{\text{SF}}^{(c)}$ or b_c is wrong.
5. **The $\langle a \rangle$ channel should measure as inert** at 573 K ($\sim 10^{-13}$ of the $\langle c \rangle$ channel). If it is not, (D.9)’s sign or $E_i^f = 3.0$ eV is wrong.
6. **Mode 2 against mode 0** on the 500 nm matched leg, interior mean, at the fitted $p_v = 1.178808$ and zero applied stress — the same protocol that isolated `loop_model`. Report N_c , c_c , $\bar{n}_c$ and d_{coarsen} .

5.3 Item (II) — the artifact is the test

1. **The boundary shell must stop diverging.** On the $1\mu\text{m}$ single crystal at 21 dpa, the surface nodes’ share of total $\langle a \rangle$ density must fall well below the measured 99.8%, and c_a/N_a there must rise above $n_{\text{iL,nuc}} = 300$. This is the sharpest available signal and it needs no new diagnostic.
2. **The domain mean must become quotable.** Interior-quartile mean and domain mean should converge toward each other; the recorded $474\times$ discrepancy at 21 dpa is the baseline.
3. **frac_kept must rise.** Re-run the transfer ledger on the 500 nm matched leg; the $\langle c \rangle$ figure of 0.96% at 1 dpa should improve, because the continuum will no longer hold loops that cannot fit. If it does not, (7) and `fits_in_crystal` disagree and one of them is wrong.
4. **Closure.** `grain_boundary_loops` must be strictly positive and monotone, and `grain_boundary` (the mobile residual) must still track `post/boundary_flux.py`’s independent integral no worse than before — on the 200 nm march, where the agreement is 3.7%. If the mobile channel degrades, loop removal is leaking into the residual.

6 Sequencing

1. **Settle the four queries of §2.8**, above all b_c . *Query 3 is blocking for item (I); item (II) can proceed in parallel, and queries 1, 2 and 4 can be settled after the machinery is in place, because each is a single constant or sign.*

2. Add the Appendix D parameter set to the 0-D; implement `mu_L/Eb_loop/C_eq_loop` with the blend, plus the table and detailed-balance unit tests. No ODE change yet.
3. Item (I) in Python, behind the `emission_model` switch, modes 0/1/2; verify mode 0 unchanged.
4. Mirror into `rate_equations_core.h`, `parameters.h` and `cpp_bridge.py`; re-check the 25-point hash.
5. Mirror into `ClusterDynamicsFEM.cpp`; the two slow steps must agree on a single-node comparison.
6. Item (II): geometry helpers, the two accumulators, state $19 \rightarrow 21$ with length-tolerant readers.
7. Item (II) in both slow steps, route G1.
8. The validation campaign of §5.
9. **Refit.** Mode 2 retires τ_{vL}^0 and E_a ; this is the fourth stale-fit item and they compound with Ω , the $\langle c \rangle$ Burgers vector and `loop_model`. Do not compare mode 2 with experiment before the refit.

7 Risks

Risk	Mitigation
b_c changes again	Blocking step 1. A move to the faulted $\frac{1}{6}\langle 20\bar{2}3 \rangle$ vector touches the same four places the $c/2$ standardization did, re-opens $f_{\text{burgers}} = 1.000000$ at the transfer, and makes $\gamma_{\text{SF}}^{(c)}$ load-bearing. Decide once, then land it as its own commit <i>before</i> item (I)
$N_{\text{cont}} = 20$ falls below the validity of (D.8)	Enforce $N_{\text{cont}} > \pi b^3/\Omega$ per family at construction; at $b_c = 0.6173$ nm that threshold is 31.8
The blend makes E_b non-smooth and the AD Jacobian discontinuous	Match value and first derivative at $N = 5$ and $N = 20$; the codebase already uses C^1 smoothsteps for exactly this reason
Emission makes the $\langle c \rangle$ system stiff near dissolution	The term is first order in c , so MoDELib3's implicit update handles it. Watch the CVODE step count, ~ 100 per point per substep today
φ^{GB} makes the boundary shell stiff	φ is C^0 by construction; promote it to a smoothstep if the failed-node count rises above today's single deterministic failure (node 41804)
Dedup efficiency falls once d_{gb} varies per node	measure the slow step before and after; if it hurts, quantize d_{gb} onto a coarse ladder — the ramp width w already sets the resolution that matters
State $19 \rightarrow 21$ breaks an old run Two stale-fit items land together	length-tolerant readers, and the length recorded in <code>summary.json</code> land items (I) and (II) as separate commits with separate validation runs, so the refit can attribute